PRACTICE PROBLEM SET 1
MTH 210
STATISTICAL COMPUTING

1. Prove that if $X \sim \text{Gamma}(\alpha, \lambda)$ and $Y \sim \text{Gamma}(\beta, \lambda)$ and they are independently distributed, then $Z = \dfrac{X}{X+Y}$ has a Beta distribution.

2. Prove that if Z has a Beta distribution, then $1 - Z$ also has a Beta distribution.

3. The negative binomial mass function with parameters (r, p) where r is a positive integer and $0 < p < 1$ is given by

$$p_j = \frac{(j-1)!}{(j-r)!(r-1)!} p^r (1-p)^{j-r}; \quad j = r, r+1, r+2, \ldots.$$

Find two different methods to generate random sample from the above probability mass function.

4. How do you generate random sample from the following probability density function

$$g(x, \theta) = \frac{\theta}{\pi(\theta^2 + x^2)}; \quad -\infty < x < \infty, \quad \theta > 0.$$

5. Find c_θ, such that

$$f(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} \le c_\theta g(x; \theta); \quad \text{for} \quad -\infty < x < \infty$$

6. Find θ so that c_θ is minimum.

7. Suppose $X \sim$ Gamma(2,3) and $Y \sim$ Gamma(3,4). Estimate $P(X < Y)$ based on simulations.

8. How do you generate random sample from the following probability density function

$$f(x) = \frac{1}{x^2}; \quad x > 1.$$

Based on simulation how do you compute $EX^{1/3}$, where X has the above probability density function.

9. A pair of fair dice are to be rolled continuously until all the possible outcomes 2, 3, ..., 12, have occurred. How do you estimate based on simulation the expected number of dice rolls that are needed.